Woodland Village Free

Offline asset manual

100 model prefabs | 3 demonstration scenes | 1 assembly prefabs

Model counts include the numbered source meshes. Assemblies and scene terrain are demonstration assets and do not increase the model count.

Table of contents

1. Installation and palette 2

2. Coordinates and modular construction 3

3. Included scenes and inventory 4

4. Modular build guide 5

1. Installation and palette

Import the unitypackage into a Unity project. The Free and Paid products use separate asset folders, so both can be installed together. Prefabs are in the Prefabs folder and demo scenes are in Scenes.

Each model prefab uses the shared ForestPalette material and ForestPalette.png atlas. Keep the material and texture with the prefabs when moving assets. The atlas uses point filtering to preserve the flat swatches.

The archive starts with the Built-in Standard material. Included Editor presets support the shared palette in Built-in, Universal Render Pipeline and High Definition Render Pipeline projects; static models need no runtime script; windmill motion uses the included small controller.

Configure your project for the chosen render pipeline first. For models, choose Tools > Woodland Village > Free or Paid > Match palette to current render pipeline. This preserves the atlas and all prefab references, and creates an editable matching material variant.

For a complete demonstration, open a scene from this product Scenes folder, then choose Tools > Woodland Village > Free or Paid > Prepare current package demo scene. The command also prepares HDRP sunlight, exposure and sky when needed. It refuses unrelated scenes and saves only package assets and the opened package demo scene.

The release is tested with Unity 6000.3.0f1, Built-in rendering, URP 17.3.0 and HDRP 17.3.0. Keep your project on a compatible installed pipeline version. HDRP demo settings are stored under Materials/SceneSettings so they remain editable.

Open a demo scene and use its Demo Camera, or drag a prefab into your own scene. Instances remain editable. Assembly prefabs are combinations of the included model prefabs; their children remain independent parts.

Static models need no runtime scripts; windmills include a removable motion controller. The palette and demo Editor menus are for normal project use. The models, assembly prefabs and scene terrain are already baked. No model generator, internal Builder or release authoring code is included.

2. Coordinates and modular construction

Unity coordinates use metres, Y up and +Z as the forward-facing direction. Model prefab roots use zero rotation and unit scale. Exported geometry must preserve the authored mounting datum; the package builder does not rotate or recenter modular geometry.

The modular construction contract uses a 2 m primary grid, 1 m secondary grid and 0.1 m detail grid. Structural wall bays rise 2.4 m. These are interface conventions, not a claim that all legacy props or completed houses snap to that grid.

Keep structural assembly children at unit scale. The authored gable trim uses model 892 at uniform scale 2 as a decorative exception; it does not add a model variant to the count. Use recorded sockets and interface measurements when connecting parts. Tile and thatch roof families have different profiles and require their own matching connectors.

The modular windmill and finished models 130/433 use the same WoodlandWindmillController with a real shaft-axis rotor pivot. Auto Start enables motion at 18 degrees per second. Call StartRotating(), StopRotating(), SetSpeed(value), or set DegreesPerSecond; signed speed reverses direction and zero holds the angle. Disable Auto Start for event-driven motion. Do not add another animation writing the rotor. This demonstrates exterior rotation, not a geared grain-milling simulation.

Legacy finished houses are standalone buildings. Modular reinterpretations are demonstration assemblies and may have different dimensions. Do not infer exact reconstruction from a similar silhouette.

3. Included scenes and inventory

FREE_Brookside_Cottage - 1237 placed instances

FREE_Copse_Crossing - 1242 placed instances

FREE_Makers_Garden - 1297 placed instances

The following counts are generated directly from this product manifest:

Asset type	Count
Numbered model prefabs	100
Demo scenes	3
Assembly prefabs	1
Scene terrain meshes	124
Shared palette materials	1

The Prefabs and Scenes folders contain the ready-to-use product inventory. No internal release manifests, generators or authoring code are included.

4. Modular build guide

Use the editable assembly prefabs as working examples. The paid product includes a cottage, a two-storey merchant house, a double-height barn and a windmill assembled from separate library models. The free kit includes the complete cottage recipe and alternative openings and joinery.

Layout and snapping

Work in metres with Unity Y up. Start with a 2 m plan grid, use 1 m infill where needed and 0.1 m increments for detail placement. Rotate square building bays in 90-degree steps. Keep model scales at one; enlarging a wall also enlarges its timber, openings and joints.

The structural wall bearing height is 2.4 m. Foundation tops in the example houses are at 0.4 m; their floor tiles add 0.2 m, so lower walls sit at 0.6 m. The merchant house upper floor bearing is 3.0 m. Build the footprint from compatible straight bays and corner pieces, then repeat upper wall rows at the next bearing level.

Model pivots use an authored bottom mounting datum. Unequal sills and overhangs do not redefine that grid. Some parts have a small mesh-to-socket offset: preserve the positions in an example assembly when swapping compatible pieces, rather than forcing every child position to a rounded value. Those offsets keep the actual openings, shafts and bearing surfaces aligned after export.

Walls, doors and windows

The new wall collection contains real empty door, window and shop-hatch openings. Inserts are separate models. A standard rectangular window socket is 1 m wide, from 0.8 to 1.8 m above the wall base. A standard door socket rises from the wall base to 2.0 m.

Select the socket wall first, then place the corresponding insert. The example cottage demonstrates a separate entrance leaf, window inserts and an open serving hatch. Retain compatible jamb widths and head heights when replacing an insert. A closed legacy window wall is a finished wall asset, not an empty socket.

Floors and access

Use plain floor tiles for bearing surfaces. The merchant example leaves a full 2 × 2 m upper-floor bay open above the stair. Its stair has a 2.4 m rise and meets the upper walking surface at 3.0 m. Keep that opening and landing clear when furnishing the interior; a decorative floor tile across the void blocks access.

Roofs

Tile and thatch are separate profile families. Match slope halves, gable fills, ridges and transition pieces within a family. The example house roof spans 4 m using four 1 m slope sections; its gables are tessellated from matching fills. Extend the longitudinal roof run with repeated matching bays. Do not connect a thatch edge to a tile edge merely because their bounding boxes touch.

The recipes demonstrate three footprints and two building heights. Finished legacy houses have individually authored proportions; these recipes are modular interpretations, not exact reconstructions of every standalone building.

Windmill

The paid windmill is built from separate tower, cap, bearing, shaft, hub, sail and mechanism parts. The moving group has one parent at the actual shaft axis; its children keep their assembled world positions. Rotate this parent, rather than spinning every individual model around its bottom pivot. The supplied WoodlandWindmillController turns at 18 degrees per second by default. The finished 130 and 433 windmill prefabs also have separate rotor children and the same controller; their zero-pose geometry, triangle winding, normals and palette UVs are preserved.

The measured shaft, hub and brake-wheel axes agree within 0.000001 m after the final datum correction. A conservative swept-rotor clearance proof and actual Unity controller and Play-mode tests are recorded in the release evidence. These checks demonstrate the rotating assembly; they do not claim a simulated grain-milling or gear train system.

Completed assemblies and scene terrain are not counted as additional numbered models.

Decorative gable trim exception

The Baker, Merchant and Barn demonstration assemblies each use one existing892_tile_gable_bargeboard_pair per gable at uniform scale2. This provides the continuous outer outline of a4m-wide gable without repeating dark trim over internal plaster tessellation seams. The trim root is placed0.12m below the eave datum so its doubled thickness aligns with the outer roof edge. Its inner cheek is placed exactly at the roof-shell end plane (rawX±halfLength), preventing a positive-area coplanar overlap between the trim top and the roof shell. This is a decorative assembly instance of the same existing model, not another counted model. Structural walls, openings, floors, roof shells and connectors remain at unit scale; the authored individual modular model profiles are unchanged. The actual matrices include the uniform scale and rotated export-origin compensation.

Windmill control from code

Select the windmill root and adjust **Degrees Per Second** and **Auto Start** on its WoodlandWindmillController component. Positive and negative speed reverse the direction; zero holds the current angle. Disable Auto Start when a game event should start the mill. The rotor keeps its authored local shaft axis and pivot. Do not add an Animation component that also writes the rotor transform.

```
using WoodlandVillagePaid;
using UnityEngine;

public class MillSwitch : MonoBehaviour
{
    [SerializeField] WoodlandWindmillController windmill;
    public void StartMill() => windmill.StartRotating();
    public void StopMill() => windmill.StopRotating();
    public void ChangeSpeed(float degreesPerSecond) => windmill.SetSpeed(degreesPerSecond);
}
```

You can also set `windmill.DegreesPerSecond = 24f` or call `windmill.StartRotating(24f)`. Each package has its own namespace (WoodlandVillageFree / WoodlandVillagePaid), so they can coexist. The free kit does not include a windmill model, but includes the reusable controller for any rotor transform you author. Removing or disabling the controller leaves static geometry. Exterior rotation is provided; grain processing and a coupled gear simulation are not.